

Q #1

Numerical Type

Award: 1

Penalty: 0

Calculus

Let f be a continuous function defined for all real numbers. Suppose

$$\int_0^3 f(x)dx = 2, \quad \int_1^2 f(x)dx = 7, \quad \int_3^6 f(x)dx = -5, \quad \text{and} \quad \int_9^{18} f(x)dx =$$

Find $\int_3^6 f\left(\frac{1}{3}t\right) dt$

Your Answer:

Correct Answer: 21

Not Attempted

Time taken: 00min 00sec

Discuss

Q #2

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

Which of the following represents the value of $\int_1^e \frac{\ln(x)}{x} dx$?

- A. 1
- B. 2
- C. 1/2
- D. e

Your Answer:

Correct Answer: C

Not Attempted

Time taken: 00min 00sec

Discuss

Q #3

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

Find $\int x e^{-3x} dx$

A. $\frac{1}{2} e^{-3x} (x - 3) + C$

B. $-\frac{1}{9} e^{-3x} (3x + 1) + C$

C. $-\frac{1}{6} x^2 e^{-3x} + C$

D. $-\frac{1}{3} e^{-3x} (x + 1) + C$

Your Answer:**Correct Answer: B****Not Attempted****Time taken: 00min 00sec****Discuss**

Q #4

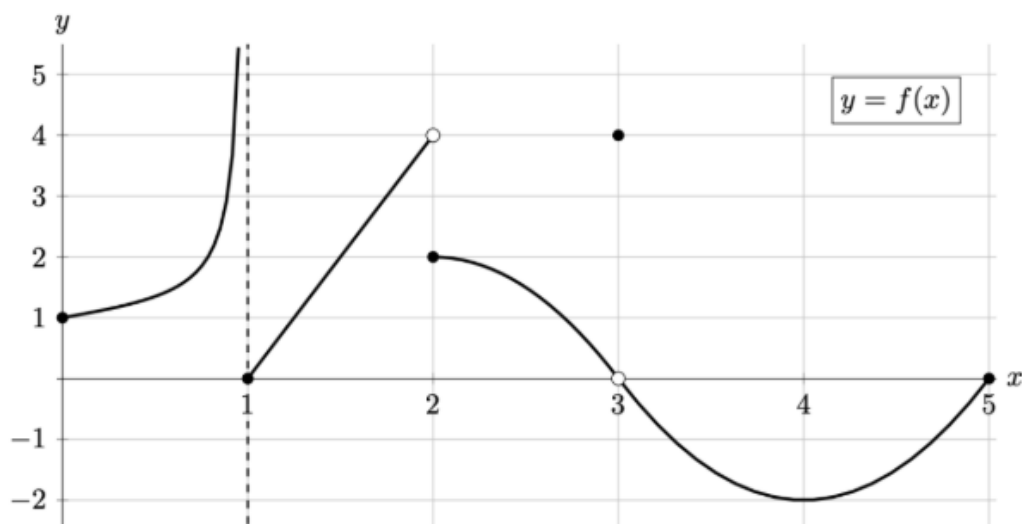
Numerical Type

Award: 1

Penalty: 0

Calculus

Below is a portion of the graph of an even function $f(x)$, which has domain $(-\infty, \infty)$ even though the graph below only shows the function on the interval $[0, 5]$. Note that $f(x)$ has a vertical asymptote at $x = 1$.



Find

$$\lim_{h \rightarrow 0} \frac{f(1.5 + h) - f(1.5)}{h}$$

Your Answer:

Correct Answer: 4

Not Attempted

Time taken: 00min 00sec

Discuss

Q #5

Multiple Select Type

Award: 1

Penalty: 0

Calculus

Some information about the derivative $p'(x)$ and the second derivative $p''(x)$ of a function $p(x)$ is provided in the table below.

x	-4	-3	-2	-1	0	1	2
$p'(x)$	1	0	-2	0	-1	0	2
$p''(x)$	-1	0	0	0	0	2	1

At which of the following values of x must $p(x)$ have a local minimum?

- A. $x = -3$
- B. $x = -2$
- C. $x = -1$
- D. $x = 1$

Your Answer:

Correct Answer: D

Not Attempted

Time taken: 00min 00sec

Discuss

Q #6

Numerical Type

Award: 1

Penalty: 0

Calculus

Consider the piecewise function

$$q(x) = \begin{cases} 7e^{x-C} + \frac{x}{3x-2} & x < 0 \\ \frac{6+5x}{2+3x+4^x} & x \geq 0 \end{cases}$$

where C is a constant. Find $\lim_{x \rightarrow \infty} q(x)$

Your Answer:

Correct Answer: 0

Not Attempted

Time taken: 00min 00sec

Discuss

Q #7

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

The function $g(x)$ is given by the equation

$$g(x) = \begin{cases} ax^2 & x \leq 1 \\ b - \ln(3x) & x > 1 \end{cases}$$

where a and b are constants.

Find the value of a such that function is differentiable at $x = 1$.

- A. $1/2$
- B. $-1/2$
- C. 2
- D. -2

Q #8

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

Consider the family of functions

$$g(x) = a \ln(x) + \frac{b}{x}$$

defined for $x > 0$, where a and b are positive constants.

Any function $g(x)$ in this family has only one critical point. In terms of a and b , what is the x -coordinate of that critical point?

- A. $x = b/a$ is a critical point where the function shows local minima
- B. $x = b/a$ is a critical point where the function shows local maxima
- C. $x = a/b$ is a critical point where the function shows local minima
- D. $x = a/b$ is a critical point where the function shows local maxima

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 00sec

Discuss

Q #9

Numerical Type

Award: 1

Penalty: 0

Calculus

The following are tables of values for two differentiable functions $f(x)$ and $g(x)$ and their derivatives. Missing values are denoted by a “?”. Assume that each of these functions is defined for all real numbers, that $f'(x)$ and $g'(x)$ are continuous.

x	0	2	3	6	9
$f(x)$	-1	?	0	-2	?
$f'(x)$	1	4	-1	?	1

x	-1	1	3	7	11
$g(x)$	-4	1	2	6	7
$g'(x)$	7	?	3	4	?

Let $z(x) = f(g(x))$. Find $z'(3)$

Q #10

Multiple Select Type

Award: 1

Penalty: 0

Calculus

Which of the following is/are TRUE?

- A. If $\lim_{x \rightarrow 5} f(x) = 0$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.
- B. If f, g , are any two functions which are continuous for all x , then $\frac{f}{g}$ is continuous for all x .
- C. It is possible that functions f and g are not continuous at a point x_0 , but $f + g$ is continuous at x_0 .
- D. If $f'(c) = 0$ then $f(x)$ has a local maximum or a local minimum at $x = c$.

Your Answer:

Correct Answer: C

Not Attempted

Time taken: 00min 00sec

Discuss

Q #11

Multiple Select Type

Award: 1

Penalty: 0

Calculus

Suppose f is a function such that $f'(x) = 4x^3$ and $f''(x) = 12x^2$.

Which of the following is /are true?

- A. f has a local maximum at $x = 0$ by the first derivative test
- B. f has a local minimum at $x = 0$ by the first derivative test
- C. f has a local maximum at $x = 0$ by the second derivative test
- D. f has a local minimum at $x = 0$ by the second derivative test

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 00sec

Discuss

Q #12

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

The derivative of $g(x) = e^{\sqrt{x}}$ is

A. $\sqrt{x}e^{\sqrt{x}-1}$

B. $2e^{\sqrt{x}}x^{-0.5}$

C. $\frac{0.5e^{\sqrt{x}}}{\sqrt{x}}$

D. $e^{\sqrt{x}}$

Your Answer:

Correct Answer: C

Not Attempted

Time taken: 00min 00sec

Discuss

Q #13

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

The function

$$f(x) = \begin{cases} e^x & \text{if } x \leq 1 \\ mx + b & \text{if } x > 1 \end{cases}$$

is continuous and differentiable at $x = 1$.

Find the value of $m - b$?

- A. e
- B. $-e$
- C. $e - 1$
- D. $1 - e$

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 00sec

Discuss

Q #14

Numerical Type

Award: 1

Penalty: 0

Calculus

Given $F(x) = (f(g(x)))^2$, $g(1) = 2$, $g'(1) = 3$, $f(2) = 4$, and $f'(2) = 5$, find $F'(1)$

Your Answer:

Correct Answer: 120

Not Attempted

Time taken: 00min 00sec

Discuss

Q #15

Multiple Choice Type

Award: 1

Penalty: 0.33

Calculus

Find the values of A and B that make

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x \geq 0 \\ A \sin x + B \cos x & \text{if } x < 0 \end{cases}$$

differentiable at $x = 0$.

- A. $A = 0, B = 1$
- B. $A = 1, B = 0$
- C. $A = 0, B = -1$
- D. $A = -1, B = 0$

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 00sec

Discuss

Q #16

Multiple Choice Type

Award: 2

Penalty: 0.67

Calculus

Is there a number b such that $\lim_{x \rightarrow -2} \frac{bx^2 + 15x + 15 + b}{x^2 + x - 2}$ exists? If so, find the value of the limit.

- A. -1
- B. -2
- C. 1
- D. There is no such b for that above limit exist

Your Answer:

Correct Answer: A

Not Attempted

Time taken: 00min 00sec

Discuss

Q #17

Numerical Type

Award: 2

Penalty: 0

Calculus

$$\lim_{x \rightarrow \infty} (x + \sin x)^{\frac{1}{x}}$$

Your Answer:

Correct Answer: 1

Not Attempted

Time taken: 00min 00sec

Discuss

Q #18

Multiple Choice Type

Award: 2

Penalty: 0.67

Calculus

If g is continuous (but not differentiable) at $x = 0$, $g(0) = 8$, and $f(x) = xg(x)$, find $f'(0)$.

- A. 0
- B. 8
- C. 1
- D. $f(x)$ is also not differentiable at $x = 0$.

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 00sec

Discuss

Q #19

Numerical Type

Award: 2

Penalty: 0

Calculus

Suppose that $f(x)$ and $g(x)$ are differentiable functions and that $h(x) = f(x)g(x)$. You are given the following table of values:

$h(1)$	24
$g(1)$	6
$f'(1)$	-2
$h'(1)$	20

Using the table, find $g'(1)$.

Your Answer:

Correct Answer: 8

Not Attempted

Time taken: 00min 00sec

Discuss

Q #20

Multiple Select Type

Award: 2

Penalty: 0

Calculus

Suppose $f(x)$ is continuous and differentiable for all x . And $-1 \leq f'(x) \leq 3$ for all x . Which of the following is/are ALWAYS true?

- A. $f(5) \leq f(3) + 6$
- B. $f(5) \geq f(3) - 2$
- C. $f(5) \leq f(3) + 10$
- D. $f(5) \geq f(3) - 10$

Your Answer:

Correct Answer: A;B;C;D

Not Attempted

Time taken: 00min 00sec

Discuss

Q #21

Numerical Type

Award: 2

Penalty: 0

Calculus

Define

$$g(x) = \begin{cases} -6x + 2 & x \leq 2 \\ x - 2 & x > 2 \end{cases}$$

Find $\int_1^4 g(x)dx$.

Your Answer:

Correct Answer: -5

Not Attempted

Time taken: 00min 00sec

Discuss

Q #22

Multiple Select Type

Award: 2

Penalty: 0

Calculus

Which of the following is/are TRUE?

A. $\int_{-5}^5 (ax^2 + bx + c) dx = 2 \int_0^5 (ax^2 + c) dx$

B. If f and g are continuous and $f(x) \geq g(x)$ for $a \leq x \leq b$, then

$$\int_a^b f(x)dx \geq \int_a^b g(x)dx$$

C. If f and g are differentiable, then

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$$

D. If f is differentiable, then $\frac{d}{dx} \sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$.

Q #23

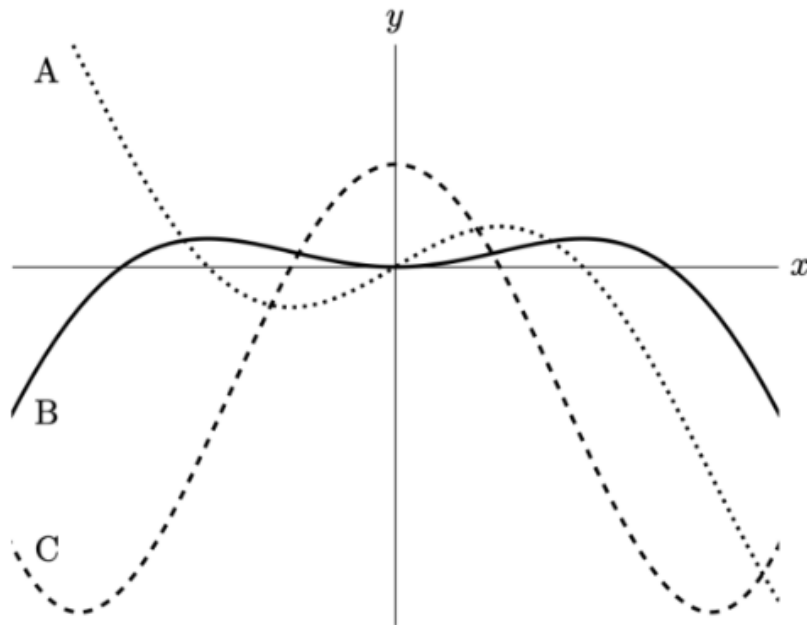
Multiple Choice Type

Award: 2

Penalty: 0.67

Calculus

Shown below are portions of the graphs of $y = f(x)$, $y = f'(x)$, and $y = f''(x)$. Determine which graph is which.



- A. $A = f(x)$, $B = f'(x)$, $C = f''(x)$
- B. $A = f'(x)$, $B = f(x)$, $C = f''(x)$
- C. $A = f''(x)$, $B = f(x)$, $C = f'(x)$
- D. $A = f'(x)$, $B = f''(x)$, $C = f(x)$

Your Answer:

Correct Answer: B

Not Attempted

Time taken: 00min 00sec

Discuss

Q #24

Multiple Select Type

Award: 2

Penalty: 0

Calculus

The function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \frac{x}{x^2 + 1} & x \leq 0 \\ ? & x > 0 \end{cases}$$

Note that the formula for $f(x)$ for $x > 0$ is unknown.

However, it is known that $f(x)$ is differentiable at each point in its domain $(-\infty, \infty)$, and that $f'(x) > 0$ for all $x \geq 0$.

Which of the following option is/are TRUE?

- A. $x = 0$ has global minima
- B. $x = 0$ has global maxima
- C. $x = -1$ has global maxima
- D. $x = -1$ has global minima

Your Answer:

Correct Answer: D

Not Attempted

Time taken: 00min 00sec

Discuss

Q #25**Multiple Select Type****Award: 2****Penalty: 0****Calculus**

Suppose $h(x)$ is a function such that $h(x)$ has exactly three critical point. Two of which are shown in the table below.

Assume that both $h(x)$ and $h'(x)$ are differentiable on $(-\infty, \infty)$

x	0	3	5	7
$h(x)$	2	?	4	4
$h'(x)$	-1	0	0	?

Further using Lagrange mean value theorem in the interval $[5, 7]$, we can determine the interval of the third critical point.

On which of the following intervals must $h(x)$ be increasing on the entire interval?

- A. $(0, 3)$
- B. $(3, 5)$
- C. $(5, 6)$
- D. $(6, 7)$

Your Answer:**Correct Answer: B****Not Attempted****Time taken: 00min 00sec****Discuss**

Q #26

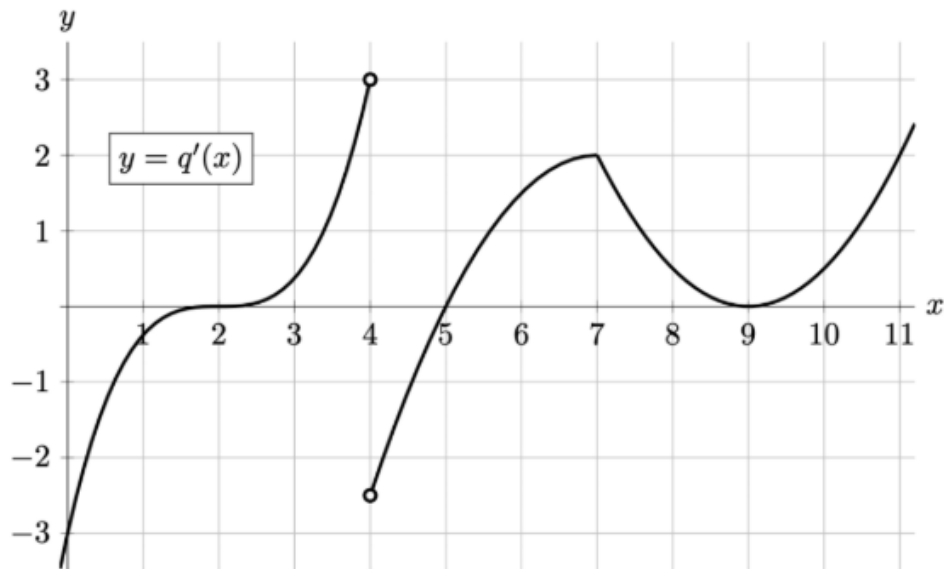
Multiple Select Type

Award: 2

Penalty: 0

Calculus

Let $q(x)$ be a continuous function which is defined for all real numbers. A portion of the graph of $q'(x)$, the derivative of $q(x)$, is shown below.



On which of the following interval(s) is $q''(x)$ positive?

- A. $(0, 2)$
- B. $(2, 4)$
- C. $(7, 9)$
- D. $(5, 7)$

Your Answer:

Correct Answer: A;B;D

Not Attempted

Time taken: 00min 03sec

Discuss

Q #27

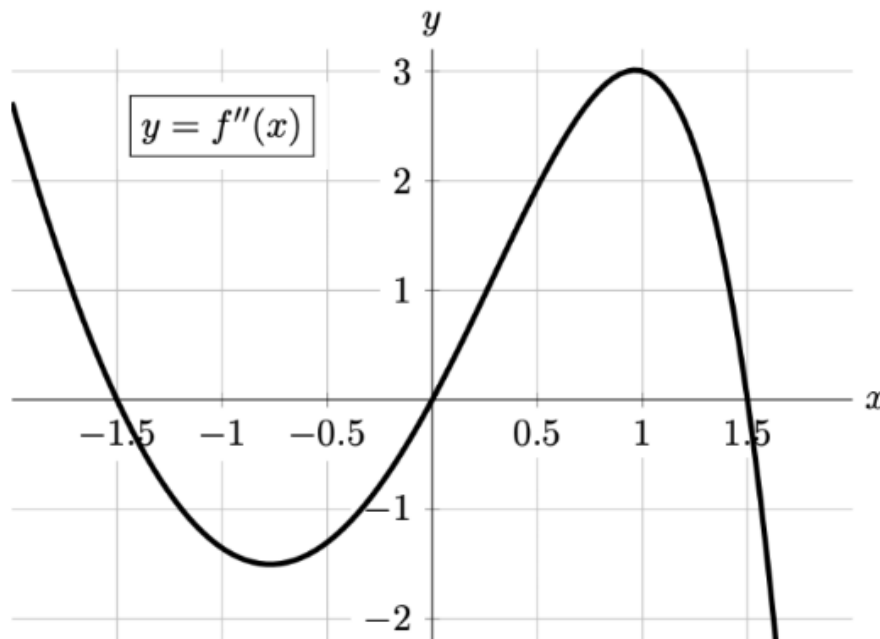
Numerical Type

Award: 2

Penalty: 0

Calculus

We consider a function $f(x)$ defined for all real numbers. We suppose that the first and second derivatives $f'(x)$ and $f''(x)$ are also defined for all real numbers. Below we show the graph of the second derivative of f . You may assume that $f''(x)$ is decreasing outside of the region shown.



Suppose that $f'(0) = 5$. How many critical points does f have?

Your Answer:

Correct Answer: 2

Not Attempted

Time taken: 00min 00sec

Discuss

Q #28

Numerical Type

Award: 2

Penalty: 0

Calculus

Consider a continuous function $f(x)$, and suppose that $f(x)$ and its first derivative $f'(x)$ are differentiable everywhere. Suppose we know the following information about $f(x)$ and its first and second derivatives.

- On the interval $(-\infty, -2)$, we have $f(x) = 2^{-x}$.
- $\lim_{x \rightarrow \infty} f(x) = 6$
- $f(2) = -5$, $f(3) = 7$, and $f(4) = 8$
- $f'(x)$ is equal to 0 at $x = -1, 2, 4$, and not at any other x -values.
- $f''(x) < 0$ on the intervals $-1 < x < 0$ and $3 < x < 5$, and not on any other interval.

Find the global minimum of $f(x)$ on $(-\infty, \infty)$? (minimum value of $f(x)$)

Your Answer:

Correct Answer: -5

Not Attempted

Time taken: 00min 00sec

Discuss

Q #29

Numerical Type

Award: 2

Penalty: 0

Calculus

If $f(x) = (1+x)(1+x^2)(1+x^3)(1+x^4)$, then $f'(0) = ?$

Your Answer:

Correct Answer: 1

Not Attempted

Time taken: 00min 00sec

Discuss

Q #30

Numerical Type

Award: 2

Penalty: 0

Calculus

Let $M = \lim_{x \rightarrow 0^+} (e^x + 3x)^{1/x}$. Find the value of $\log_e M$.